

Framework to Facilitate Partnership Agreement between CGIAR and the Kenya Plant Health Inspectorate Service (KEPHIS) concerning the Root, Tuber and Banana - East Africa Germplasm Exchange Laboratory (RTB EAGEL)

Background

Rate limiting steps in breeding pipelines of root, tuber and banana (RTB) crops (*Musa*, Cassava, Sweetpotato, potato, yam) and their scaling to seed systems frequently involve limited access to virus-free plantlets of priority genotypes which can be distributed, particularly for multi-locational trials.

IITA and CIP of the CGIAR have collaborated closely with KEPHIS for more than 20 years, around germplasm health issues of RTB crops. To date, they have utilized existing laboratory facilities and have contributed equipment and screen houses in addition to capacity building. As activities have expanded, and the demand for quality germplasm of RTB crops has escalated, in parallel with the ageing of existing facilities, there is an urgent need to upgrade the facilities. IITA and CIP have obtained resources from GIZ through a fund called Crops to End Hunger (CtEH) administered by the Breeding Resources Initiative of the OneCGIAR, to support the building of a new laboratory, called here the 'Root, Tuber and Banana East Africa Germplasm Exchange Laboratory (RTB EAGEL). This investment is time-bound from 20th June 2023 to 31st December 2024. The laboratory will be a regional hub for the acquisition and distribution of quality planting material of RTB crops to breeding programs and seed entrepreneurs in the region. The main clients will be the National Agricultural Research Institutes' (NARIs) breeding programs and seed entrepreneurs wanting to introduce new varieties. It will also be a centre of expertise for training researchers and industry practitioners.

This document attempts to identify the main areas of partnership that need discussion and agreement to ensure the continuation of a productive and healthy relationship between KEPHIS and the CGIAR, hereby represented by IITA and CIP, and that the laboratory fulfils its intended purpose in a sustainable manner. The project was initiated in June 2023, should be completed by December 2024 and currently includes construction and equipping of the laboratory (see Annex 1 for details). Attempts will be made to secure funding to increase the client base, including exposing NARIs and RTB Crop Networks to the services and opportunities offered by the RTB-EAGEL, although this is in no way guaranteed.

Purpose and role of RTB EAGEL

A centralized facility is proposed, within KEPHIS Plant Biosafety and Quarantine Station (PBQS), Muguga, that can operate at scale and provide services reliably to CGIAR/NARES/private sector and RTB crop networks and serve as a center of excellence and training.

Services to be provided

1. Pathogen detection including virus indexing using state-of-the-art diagnostics

2. Phytosanitation, including virus elimination
3. *In vitro* germplasm conservation of high-priority genotypes
4. Low-density and low-throughput DNA fingerprinting for genotyping and true-to-type determination
5. *In vitro* germplasm multiplication and distribution
6. Training
7. Hosting of relevant research projects

Investment by CGIAR through IITA and CIP

The CGIAR, through IITA and CIP, are investing in building a state-of-the-art laboratory with adequate equipment and trained personnel to provide the required services, as listed above. IITA and CIP will be providing further training, initial consumables, data management system and a business plan to guide the operationalization of the lab in a sustainable manner (see Annex 1).

Investment by KEPHIS

KEPHIS will provide the land for the RTB-EAGEL and basic utilities ie. electricity, water, sewage, telephone, internet etc. and services among others, and will contribute, pay and maintain competent staff to run the facility.

Areas for discussion

Definition of terms:

Client: those who keep long-term and on-going relationships, such as CGIAR, NARIs

Customers: Customers are transactional-based relationships with a business activity such as seed entrepreneurs.

Scope and Nature of the Partnership:

1. At the core of the partnership between KEPHIS, IITA and CIP will be mutual respect and equality. This is an equal partnership. No one entity should seek to dominate or exclude another partner, but rather foster a cordial, positive and constructive partnership serving the interests of RTBs in the region.
2. All promotional material or communication related to RTB EAGEL should recognize KEPHIS, CGIAR, IITA and CIP.
3. For purposes of branding, profile and attracting customers and clients, the lab will be named “Roots, Tubers and Bananas East Africa Germplasm Exchange Laboratory” (RTB EAGEL) which was agreed through the poll conducted in March 2024 in which 51 people from KEPHIS, CIP and IITA were surveyed for their opinions.
4. RTB EAGEL will be branded in a non-partisan way.
5. The Partnership Agreement should be CG-wide, and not limited to particular crops or Institutions.

Management:

1. Strategic, management and access decisions would be taken by a Management Committee (MC). This includes decision on operations and determining fees for services to be charged to the clients.
2. The Management Committee (MC) would be composed of two representatives from KEPHIS, two representatives from IITA and two representatives from CIP. MC members will be nominated by the Institute they represent. A minimum quorum of four people, with at least one representative from each organization, would be required either in person or online, for any decisions to be passed.

Duration of Agreement: 5 years renewable.

Ownership:

1. KEPHIS would own the laboratory as it is built on KEPHIS land.
2. Equipment would be owned by the CGIAR with a view to donating it to KEPHIS after the end of this agreement.
3. Any relocation of equipment would have to be agreed by the MC

Ownership of germplasm:

1. As a service Center, KEPHIS will not own any intellectual property governing germplasm.
2. Organizations supplying material will decide the type of services required and distribution decisions.
3. Institutes owning the intellectual property of germplasm will be responsible for signing 'Material Transfer Agreements (MTA)'.

Maintenance:

1. KEPHIS commit to maintain the facilities for the duration of the agreement.
2. Maintenance and improvement of the laboratory structure would be financed and factored into a revolving fund. During the ramp-up period of (5-6 years) if funds are insufficient, maintenance of equipment will be the responsibility of CGIAR.

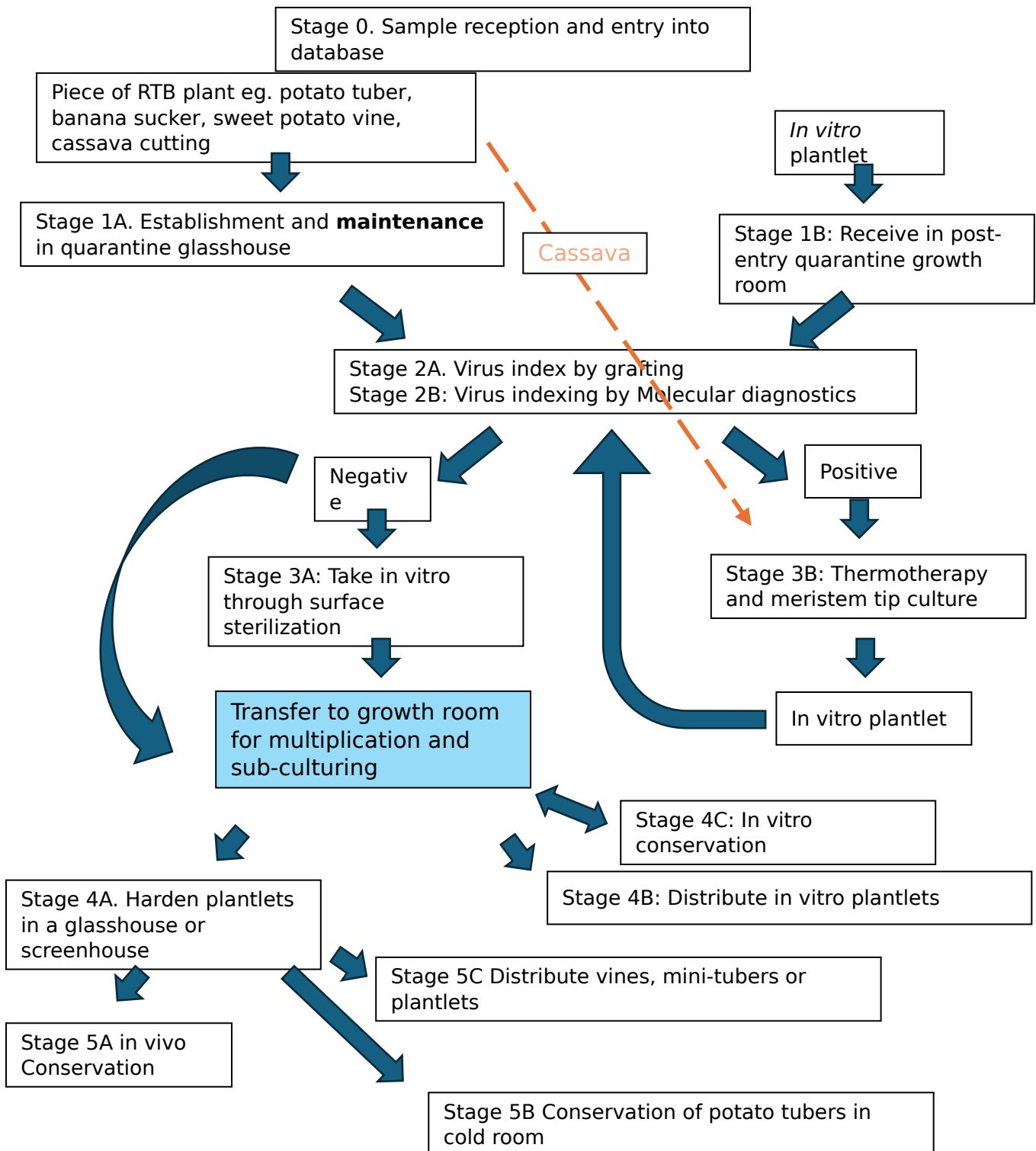
Access

4. Access decisions would be taken by a Management Committee (MC). This includes decisions on operations and determining fees for services to be charged to the clients and customers. The MC may consider different rates for CGIAR, non-CGIAR NARES/non-profit organizations, and private sector/profit-organizations.
5. CGIAR projects would be hosted as a priority at a fee determined by the business plan's current iteration, which will be updated regularly and agreed upon by the Management Committee

Staffing

6. KEPHIS will provide, pay and maintain minimum staffing levels as outlined in the organogram (Annex 2), to provide the services offered.
7. CGIAR would be able to second their own staff to the facility at no additional cost between CIP, IITA or KEPHIS.

Overall Generic Workflow



Project Synopsis:

Challenge & Solutions:

Challenge	Solution	BPAT recommended and/or One CGIAR priority
Current efforts in East Africa to conduct virus indexing, virus elimination, maintenance, propagation and distribution in vegetatively propagated crops (VPC) are fragmented across CGIAR and NARES. Duplication of efforts and reduced scale are expensive and unsustainable.	A centralized facility is proposed, within a NARES, that can operate at scale and provide these services reliably, cost-effectively and sustainably to CG/NARES crop networks.	This addresses One CGIAR priorities by working across centers (CIP and IITA), embedded within a NARES (KEPHIS). The need for CG / NARES engagement to effectively run several sweet potato breeding programs in ESC Africa was recognized by BPAT. The need to connect and collaborate across hubs including NARES was also highlighted in the cassava BPAT. The sweet potato BPAT also recognized that inadequate storage space, screen houses, power supply, labs and office space represent a considerable challenge for CIP. In addition, “It seems that many basic infrastructure needs are not met in core funding and are not comprehensively addressed through project funding”. BPAT recommended that tissue culture capacity be increased for potato in East Africa to drive greater efficiencies in the breeding program.
Due to limited capacity and facilities for phytosanitary cleaning and testing in the region there is limited access to virus free plantlets of priority genotypes of VPC which can be distributed,	Phytosanitary cleaning and testing of VPCs is a knowledge and capital-intensive process that would require huge investment to implement and maintain in each individual country. A	This initiative would help to address the recommendation in the potato BPAT to improve the standard of testing operations within the NARES/CG network. In addition, the

particularly to target population of environments (TPE) for multi-environment trials (MET). This results in delays in selection of appropriate genotypes and seed multiplication of released varieties.	centralized facility operating at scale would be more cost effective and efficient. It would acquire, clean-up, maintain and distribute priority breeding material and released varieties of VPC in a pathogen free state to supply MET and feed the seed systems on a regional basis.	sweet potato BPAT states “The responsibility for the provision of healthy planting material of elite varieties needs to be clarified. NARES typically have the mandate to release new varieties and should spearhead this and provide the necessary support”. The cassava BPAT recommends increased yield testing across TPE within-country and across Africa (broad vs specific adaptation). By facilitating movement of germplasm, the KEPHIS hub supports this goal. The banana BPAT highlights the inadequate supply of planting material to farmers of new released varieties as a constraint.
Implementation of quality control, protocols, systems and capacity associated with clone identity, maintenance, virus elimination and indexing, rapid propagation and distribution of VPC is variable in the region, increasing chances of errors that set back breeding progress and can lead to dissemination of infected planting materials. Current KASP fingerprinting services require too large a scale for local purposes.	The VPC facility will serve as a center of excellence and training for international SOPs for the region. It will run KASP markers at appropriate scale and curate a database of genetic fingerprints of priority clones for quality control and maintain clones in a virus free state for distribution. This will stimulate acceleration of genetic gains per dollar spent, and modernization of VPC breeding and increase capacity of regional institutions managing VPCs.	This addresses the BPAT recommendation to streamline testing operations within the NARES/CG network in East Africa for both potato and sweetpotato.
The pipeline for supply and distribution of pathogen free planting material of VPC has not been properly costed and incorporated into a business model with	Support KEPHIS to develop a business plan and service agreements and sensitize breeding networks to the services available	

proper cost recoveries to ensure sustainability		
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Objectives and Targets:

Objective	Target	Method & Source- please explain how target will be measured, by when and relevant source/ evidence
Establish an energy-efficient, streamlined laboratory, for delivering on service agreements for virus elimination, indexing, in vitro maintenance, rapid propagation and regional distribution of VPC at Kenya Plant Health Inspectorate Service Plant Quarantine and Biosecurity Station (KEPHIS-PQBS) at Muguga, Nairobi. This will enable more rapid and efficient regional breeding and release of new varieties of cassava, sweetpotato, potato, Musa and yam.	The facility will be tested with ten clones of each of the target crops processed using the new laboratory and systems to confirm they meet expected standards. At least 70% of the entire new facility powered through solar energy.	Results of the pilot study will be assessed to forecast achievements in future years. The ten target clones are fingerprinted and tracked in the database, including through the phytosanitary cleaning and testing processes and are demonstrated to be virus free and true to type by Dec. 2024. Electricity consumption will be recorded, and the proportion contributed by solar power calculated.
Strengthen protocols, systems and capacity associated with sample tracking, maintenance, virus elimination and indexing, rapid propagation and distribution of VPC to reduce errors that lead to breeding delays and risk of dissemination of infected or mislabeled materials.	The same clones per target crop as above processed using new sample tracking procedures including fingerprinting, following optimum SOPs for virus indexing and elimination, by empowered KEPHIS technicians, at least 40% of whom are women.	SOPs used at CIP/IITA will be revised and adjusted to KEPHIS-PQBS context and training provided for their implementation, which will subsequently be recorded through tracking LIMS by Dec. 2024.
Build sufficient capacity and systems to enable KEPHIS to maximize financial sustainability.	Demonstrate an appropriate business model and systems by processing ten pilot clones per crop using service agreements with cost-recoveries by the end of the project.	The percentage of cost recovery with implementation of the financial sustainability plan will be monitored and reported.

Project Implementation:

This project will be implemented in two phases. Phase 1 (01-2023 to 12-2024) will focus on establishing facilities, systems and procedures, including training of KEPHIS staff ('training the trainers'). At the end of Phase 1 the facilities and systems will be tested to ensure the

required performance and outcomes will be achieved in Phase 2. Phase 2 (01-2025 to 01-2027) which is outside the current funding and independent, will focus on the consolidation of crop networks, extend the capacity sharing to the CG-NARES network and distribution of breeding material and varieties to entities within the networks. An implementation coordinator (IC) will oversee the day-to-day implementation of the project and be responsible for time and financial management and liaising with all partners.

Objective 1 Infrastructure: Establish an energy-efficient, streamlined laboratory, for delivering on service agreements for virus elimination, indexing, *in vitro* maintenance, propagation and distribution of VPC

1.1. Upgrade central water and electricity systems to support sustainability.

KEPHIS have committed to upgrade the borehole pump using complementary funding to ensure water security at KEPHIS. A number of large underground water tanks used for water harvesting are likely leaking. Cost effective options for water harvesting are to repair the tanks and to install additional tanks to harvest water from the glasshouse and building roofs. Solar pumps would also dramatically reduce energy requirements from borehole pumps contributing to the outcome of climate resilience.

Milestone: Current water and electricity systems upgraded by Dec 2024.

1.2 Rebuild aging in vitro and inadequate molecular laboratory infrastructure

A new building will house all necessary infrastructure and equipment for service-oriented virus elimination, testing, multiplication and delivery of pathogen free planting material of VPC, in contrast to the current facility in which infrastructure are inadequate, unreliable and unsterile. An initial needs assessment has indicated a per annum requirement of a minimum of 6,250 plantlets multiplied through tissue culture, 450 genotypes for virus elimination, 4,050 samples for indexing, with approximately 670 genotypes maintained routinely each with multiple copies. The old building will remain and be refurbished to some extent and be used for training and other crops. We will aim to cover 70% of the monthly lab energy requirement by solar power, saving ~3500 USD/month. Modular solar units will enable expansion of the system over time as resources become available. The new building design will be subject to external review including gender appropriateness for improvements by March 2023. Ergonomic equipment will be purchased where possible including seating for laminar flow cabinets and pipettes.

Once the equipment is installed it will be officially handed over to KEPHIS. We will seek to negotiate an equipment maintenance contract with the International Livestock Research Institute (ILRI) that has a team of maintenance engineers. Where applicable maintenance contracts will be agreed with manufacturing companies by June 2024, if representatives are available in Kenya. A lab manager (LM) will initially be funded by Seed Equal although that person will transition to KEPHIS as the project progresses.

Milestone: The build will be complete by March 2024 and installation of equipment complete by June 2024.

1.3. Upgrade at least one glasshouse facility with more effective temperature control for optimum propagation

The temperature in the current glasshouses is not well controlled and depending on the time of year can reach over 35°C and fall to below 10°C. This is not conducive for hardening of plantlets or consistent plant growth. Limited upgrade of the temperature control of a few chambers in existing glasshouses will be implemented

Milestone: Complete by Dec. 2024.

1.4 Renovate common use and service areas for improved working conditions

A number of existing buildings require some renovation to enhance safety, security, health and the environment (SSHE). This includes the KEPHIS caretaker lodgings, canteen upgrade and renovation of service building to create rest rooms for those in need, including pregnant or breastfeeding women

Milestone: Caretaker lodgings, canteen upgrade and provision of rest room complete by June 2023.

Objective 2 Capacity Sharing: Strengthen protocols, systems and capacity to develop, maintain and multiply virus free germplasm for regional distribution

2.1 Implement a bar-coded clone tracking and data management system including an on-line application system for germplasm request and tracking of progress and distributions
Best-bet options for a clone tracking and data management system will be assessed through discussion with BRI, Germplasm Health Units at IITA and CIP, supported by CIP and IITA resident capacity in Nairobi. Funds are included for system customization to the specific needs of the facility. Basic ICT infrastructure will be purchased for the lab to implement these systems, including basic networking infrastructure, computers and accessories, barcode printers, barcode scanners, tablets and others (orders started by Sept. 2023). These systems will minimize errors in tracking planting material that could lead to breeding delays and transfer of erroneous material.

Milestone: Best-bet options for a clone tracking and data management system complete by June 2023 and operational by Sept. 2024

2.2 Establish SOPs and training for phytosanitary cleaning, disease indexing and traceability

The facility will be staffed by KEPHIS with at least 40% women, with capacity strengthening, using hybrid learning approaches provided by IITA and CIP, including the use of best practices and standard operating procedures established for the safe distribution of virus-free planting materials developed by CGIAR GHUs. SOPs used at CIP/IITA will be revised and adjusted/uniformized to KEPHIS-PQBS context (complete by Sept. 2023) and will be reviewed by woman to identify gender implications. Training will be provided for their implementation, at first by visit of KEPHIS technicians to IITA and CIP Genebanks to receive training and witness standard operations in high quality environment (completed by June 2024). Subsequently backstopping (virtual and physical) from CIP/IITA Genebank staff to support implementation at the new facilities in Nairobi will take place.

Milestone: KEPHIS staff confidently implementing best practices and SOPs by Dec. 2024.

Genotyping will be needed on a limited number of samples, but at regular intervals, for quality control (QC) purposes, with results being made available within days. A small-scale KASP system will be developed for this purpose. KASP does not require any additional equipment than what is routinely available in any molecular lab. This would involve purchase of primers and probes, and transfer of capacity.

Milestone: KASP operational by Dec. 2024.

Objective 3 Financial Sustainability: Build sufficient capacity and systems to enable KEPHIS to maximize financial sustainability

The potential business model(s) will be identified, and financial feasibility studies will be carried out to understand net present value and internal rate of return for the investment made in selected business model(s). During this process, we will refine the revenue generation model by strengthening pricing strategies and cost management. Further, financial management can be strengthened by looking at cash flow, balance sheet and

income statement. This will enable us to execute selected potential business models successfully.

- 3.1. Identify and develop potential sustainable, scalable business model(s)
- 3.2. Financial and economic cost-benefit analysis
- 3.3 Identify macro-environment external factors in the business operation led by KEPHIS
- 3.4. Strategies for strengthening potential business models and partnerships

A business consultant will be engaged through the Breeding Resource Initiative (BRI) to assist with business plans and service agreements. All services will be fully costed to guide full cost-recovery and contribute to sustainability

Project Benefits:

The project will enable pathogen-free quality breeding material and varieties to be distributed to CGIAR and NARES breeding initiatives throughout ESC Africa. This will facilitate multi-environment trials (METs) for more accurate selection for broad or specific adaptation. Importantly it will readily provide quality breeding material to those NARES that are purely engaged in mid to late-stage testing and selection. The faster that NARES receive new breeding lines for testing, the more likely it is that selected new varieties will be released more quickly and have timely impact with farmers. The more NARES that are engaged, the larger the footprint of impact. The facility will also feed quality planting material into propagation facilities regionally.

Market segment and Beneficiaries:

Twelve market segments have been identified. The below table highlights the largest one for each crop. The full table is provided in Annex A.

Market Segment	Million hectares	Number of product users or beneficiaries
<u>Potato</u> : ESA, EAF, Table, Red, Pink & Cream, African Highlands, Short dormancy, Early	Kenya (200K ha), Rwanda (104K), Ethiopia (72K), Uganda (36K) = 412K ha	It is estimated there are ~187million people in the market segment of which ~63.6 million are considered to live in poverty with ~42.3 million are under-nourished. There are ~ 1.1m people in the market segment footprint ie. living on the hectares where the crop is grown, and of these nearly half a million are living in poverty and nearly 328K are under-nourished.
<u>Sweet potato</u> : ESA, EAF, Food (Home consumption & fresh markets), White, Rainfed, Early/Intermediate	Tanzania (500K ha), Uganda (450K), Ethiopia (60K), Rwanda (180K), Kenya (80K), Burundi (90K), Eastern DRC (110K) = 1,470,000 ha	It is estimated there are ~288 million people in the market segment of which ~137 million are considered to live in poverty with ~69 million under-nourished. There are ~ 3.4m people in the market segment footprint ie. living on the hectares where the crop is grown, and of these ~1.8m are living in poverty and nearly 1m are under-nourished.
<u>Cassava</u> : Fresh for East, Southern and Central Africa; Highlands/mid altitude/coastal.	Tanzania (832Kha), Uganda (1,008Kha), Rwanda (171Kha) = 2,010,000ha	It is estimated there are ~90million people in the market segment of which ~42.5 million are considered to live in poverty with ~21.6million under-nourished. There are ~ 4.2m people in the market segment footprint ie. living on the hectares where the crop is grown, and of these ~1.9m are living in poverty and nearly 1m are under-nourished.
<u>Musa</u> : Highland Cooking banana: Matoke (3x) (EA)	Uganda (550Kha); DRC (110Kha), Rwanda (236Kha), Burundi (184Kha), Kenya (58Kha) = 1,138,000ha	It is estimated that there are ~140 million people in the market segment of which 88m are considered to live in poverty with ~1.8m under-nourished. There are ~ 3.8m people in the market segment footprint ie. living on the hectares where the crop is grown, and of these ~1.9m are living in poverty and nearly 0.5m are under-nourished.
<u>Yam</u> : White and water yam adapted to low and mid-altitude for boiled consumption	East and central Africa (220 Kha)	Demand for yam as a nutritional and income security crop is increasing in East Africa and One CGIAR is starting to engage in this market segment. By supplying disease-free planting material to partner projects, it is estimated that at least 50,000 smallholder

		households will benefit from accessing and growing improved yam cultivars.
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Project Outcomes:

Immediate outcomes:

Category of outcome	Details
Regional coordination and efficiencies in the CGIAR-NARES breeding network e.g. number of breeding lines and organizations benefitting	Each of the four main crops, and the 12 market segments benefiting from this project will interact with a CGIAR-NARES breeding network of at least five NARES, providing them with quality planting material of new varieties and breeding lines for mid to late-stage testing. The facility would form a hub for regional crop network coordination.
On-site climate change adaptation or reduced emissions e.g. use of solar power, water efficiency	The lab building will be built with a suitable roof to accommodate 750m ² of solar panels. At least 70% of the power of the new build will be derived from solar energy. This will be based on lithium batteries with an estimated lifespan of 15 years and a 10 years' warranty. Following assessment of the water situation at KEPHIS, existing rainwater catchment tanks may be repaired.
Other on-site environmental benefits e.g. soil erosion	An aesthetic outdoor area adjoining the staff canteen will be upgraded for the relaxation and outdoor health benefits of staff.
Staff capacity e.g. skills and knowledge, number trained (gender-disaggregated)	At least one technical personnel working on each crop will be trained in quality management and SOPs for clonal multiplication, phytosanitation and disease indexing. At least 40% of those trained will be female.
Min. 30% more women in leadership positions	Head of the KEPHIS PQS at Muguga is a woman. Efforts will be made to recruit a female project implementation manager. Efforts will be made to develop leadership skills and promote women as KEPHIS lab managers, or crop leads.
Labor and job safety e.g. reduction of manual labor, workplace wellbeing (gender-disaggregated)	Current tissue culture facilities at KEPHIS have clear health concerns and do not provide a safe, conducive environment for work. New facilities will provide a working environment with natural sunlight, ergonomic chairs and pipettes and effective air quality control. An existing building next to the new facility provides a rest area and catering, but will be upgraded.
Gender responsive facilities and equipment	We will adopt a gender responsive approach to plan and design new facilities and choose equipment. This includes renovation and provision of a rest room in an existing facility for pregnant or breastfeeding women, and ergonomically designed chairs and other equipment in the lab area.

Accelerated breeding e.g. number of generations per year	By coupling virus cleaning/indexing with rapid multiplication and dissemination, this project will accelerate breeding by enabling the faster distribution of relatively early-stage breeding clones (eg. PYT) to a broader range of TPEs for METs and on-farm trials (OFTs). This will not only shorten the time it takes to release a variety but also increase the number of countries/regions where the potential varieties will be tested and simultaneously released.
Genetic gain e.g. through selection intensity	Using quality planting material in METs or OFTs, can effectively increase heritabilities. By facilitating a greater number of MET and OFT in a larger number of TPE, more quickly, and using quality planting material, genetic accuracy of advancement decisions for particular TPEs could be improved, thereby effectively increasing genetic gain, and information pertaining to genetic gain. This would be achieved during an independent Phase II of the project and would be a legacy outcome of this Phase.
Research expansion e.g. more plants within greenhouse	During Phase II, 30% more MLTs will be conducted in the region, across crops. Triple the number of <i>in vitro</i> plantlets of improved varieties will be available for distribution. Tracking and documentation systems will be in place clearly communicating which improved varieties are available for distribution regionally (considering varietal release status). SME that multiply banana will start with clean, virus indexed material. This would be a legacy outcome of Phase I.
Research quality e.g. improved data quality	The availability of sufficient disease-free quality standardized planting material of confirmed identity of specific varieties will improve the data quality of MLTs in TPEs and OFTs.
Trait assessment e.g. ability to measure or phenotype	Making quality planting material available enables more extensive phenotyping resulting in more data points.
Cost savings by the applicant breeding center e.g. forecast future savings	By going to MET early, the breeding programs will save considerable resources that would have otherwise been used to run successive years of limited on-station trials.
Cost savings by other networked breeding programs	Breeding programs in the crop networks will save resources by evaluating breeding lines derived from quality planting material. Time and resources will not be wasted trying to access quality planting material. Through co-ordination of breeding programs in networks, data from across METs will be shared, enabling decisions regarding selections to be made with greater accuracy and earlier, saving resources.
Financial sustainability (include when and how financial sustainability will be achieved beyond the duration of the project)	Through establishment and implementation of appropriate business models, services such as virus elimination, virus indexing and distribution of pathogen free planting material will be charged on a cost-recovery basis. With over 70% of power being derived from solar, these services will be energy efficient and thus cost-recovery fees will be minimized and

	likely more acceptable to users. This will contribute to financial sustainability of the service.
Equipment and facility maintenance (include how maintenance will be sustained beyond the duration of the project)	We will seek to negotiate an equipment maintenance contract with ILRI that has a team of maintenance engineers. Where applicable a maintenance contract will be agreed with the manufacturing company of specific equipment, if representatives are available in Kenya. As cost recovery mechanisms are put in place, building and equipment maintenance will be budgeted and accounted for.
Improved risk management	Through the provision of a new purpose-built facility, health and safety and working conditions will be dramatically improved. The risk of contamination of tissue culture products will be drastically reduced. Improved virus indexing facilities will reduce the risk of spreading disease. Digital tracking systems and genetic fingerprinting will reduce the risk of variety mixes and mislabelling.

